

A recessive mutation in a gene called *FANCN* is suspected to increase cancer risk. A study finds that the frequency of the wild-type dominant *FANCN* allele in an Australian population is 0.98. Assuming the population is in Hardy-Weinberg equilibrium, the frequency of individuals homozygous for the recessive *FANCN* mutation in this population is approximately:

- ✔ ☒ A. 0.0004. [57%]
- ☐ B. 0.02. [31%]
- ☐ C. 0.039. [8%]
- ☐ D. 0.96. [2%]

Correct

57% answered correctly

## Explanation:

### Hardy-Weinberg equilibrium

#### 1. Determine alleles & possible genotypes

| Alleles                              | Possible genotypes        |
|--------------------------------------|---------------------------|
| F (wild-type dominant <i>FANCN</i> ) | FF (homozygous dominant)  |
| f (mutant recessive <i>FANCN</i> )   | Ff (heterozygous)         |
|                                      | ff (homozygous recessive) |

#### 2. Use HWE equations to derive frequency of ff

| Allele frequency using<br>HWE equation 1     | Genotype frequency using<br>HWE equation 2                                    |
|----------------------------------------------|-------------------------------------------------------------------------------|
| $p + q = 1$                                  | $p^2 + 2pq + q^2 = 1$                                                         |
| $p$ = frequency of F<br>$q$ = frequency of f | $p^2$ = frequency of FF<br>$2pq$ = frequency of Ff<br>$q^2$ = frequency of ff |

#### 3. Use given information to calculate $q^2$

- Given that  $p = 0.98$  solve for  $q$  using  $p + q = 1$
- Use  $q^2$  to calculate the frequency of ff

HWE = Hardy-Weinberg equilibrium

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When populations are in [Hardy-Weinberg equilibrium](#), two equations can be used to predict [allele](#) and genotype frequencies:

- $p + q = 1$ : Applies to a gene with two alleles (F and f), where  $p$  represents the frequency of one allele (F) and  $q$  represents the frequency of the other allele (f).
- $p^2 + 2pq + q^2 = 1$ : Can be used to calculate genotype frequencies (ie,

frequencies of FF, Ff, and ff) from the allele frequencies. Here,  $p^2$  and  $q^2$  represent the frequencies of the two **homozygous** genotypes FF and ff, respectively. The heterozygous genotype (Ff) frequency is represented by  $2pq$ .

In this question, a **recessive mutation** in the *FANCN* gene is suspected to increase the risk of **cancer**. In an Australian population, the frequency of the wild-type dominant *FANCN* allele is **0.98**. Assuming that this population is in **Hardy-Weinberg equilibrium**, the frequency of this dominant allele can be used to calculate the **frequency** of the **recessive mutant allele** as well the frequency of individuals **homozygous for this mutation (ff)**:

1. Solve for  $q$ . Let  $p$  represent the frequency of the wild-type dominant *FANCN* allele (F) and  $q$  represent the frequency of the mutant allele (f):

$$p + q = 1$$

$$0.98 + q = 1$$

$$q = 1 - 0.98 = 0.02 \text{ (Choice B)}$$

2. Calculate the frequency of the homozygous recessive genotype (ff):

$$\begin{aligned} \text{Frequency of homozygous recessive genotype} &= q^2 = 0.02^2 \\ &= 0.0004 \end{aligned}$$

Therefore, the frequency of individuals in this population that are homozygous for this mutation is approximately **0.0004**.

**(Choice C)** **0.039** is the approximate frequency of heterozygotes in the population ( $2pq$ ).

**(Choice D)** **0.96** is the approximate frequency of individuals homozygous for the dominant wild-type allele ( $p^2$ ).

### Educational objective:

The Hardy-Weinberg equations are used to relate allele frequencies to genotype frequencies. For populations in Hardy-Weinberg equilibrium, expected genotype frequencies are calculated using allele frequencies.